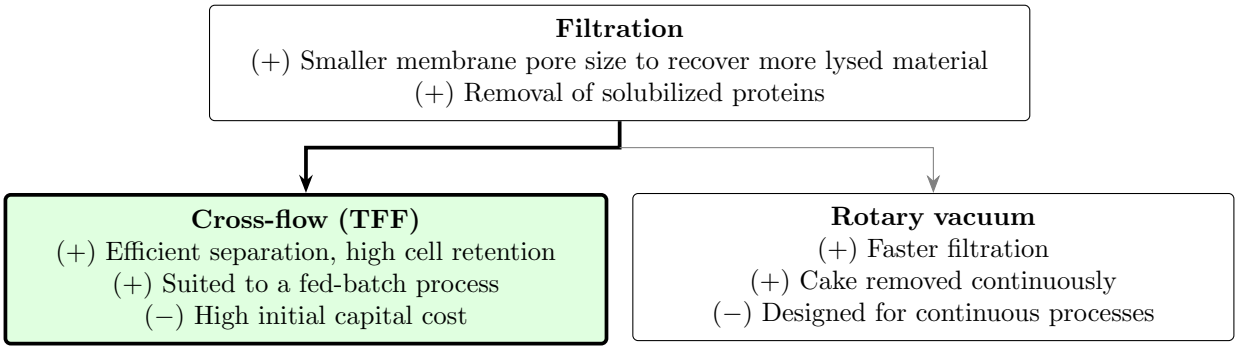
Microfiltration (TFF): Concentration and Diafiltration

UBC CHBE capstone · Montbretin A production · PFD 400: T-402, P-406, F-401, M-402

Daniel Olaniyan

Equipment selection



Selected option shaded. Cross-flow filtration (F-401) fits the batch-wise downstream train and retains the cell debris; the capital cost is the trade-off. Adapted from the capstone report, Figure A3.

Operation

The homogenized lysate is filtered in two stages through a 0.2 µm tangential-flow membrane. The product, Montbretin A (MbA), passes into the permeate; retained biomass returns to the feed tank in the retentate.

1. **Concentration mode (constant flux).** Part of the feed passes the membrane as permeate and the rest returns to the tank. Some retained biomass builds a cake on the membrane. As the feed volume falls, the retained biomass concentrates, the cake resistance grows, and TMP is raised to hold the permeate flow constant, until the membrane's maximum TMP is reached.
2. **Diafiltration mode (constant TMP).** Once maximum TMP is reached, buffer is added to the tank at the permeate rate, holding the feed volume constant. The cake resistance, TMP and permeate flow then stay constant. The stage ends when the required fraction of MbA has been washed into the permeate.

Given

Permeate flow	Q_P	$1.5\,\mathrm{m^3/h}$
Maximum TMP	$\mathrm{TMP_{max}}$	$4\,\mathrm{bar}$
Initial feed volume	V_F°	$4.8\,\mathrm{m^3}$
Initial retained-species concentration in feed	C_F°	$27.834\,\mathrm{g/L} = 27.834\,\mathrm{kg/m^3}$
Membrane area	A_m	$4 \times 80\,\mathrm{m^2} = 320\,\mathrm{m^2}$
Feed viscosity	μ	$0.001\,\mathrm{Pa\,s}$
Specific cake resistance	R_c^*	$1 \times 10^{11}\,\mathrm{m/kg}$ (assumed)
Membrane resistance	R_m	$1 \times 10^{9}/\mathrm{m}$ (assumed)
Diafiltration target	C_{FP}/C_{FP}°	0.05 (5 % of MbA left in the retentate)

Find

(1) Operating time in concentration mode, to reach $\mathrm{TMP}_{\max}$; (2) operating time in diafiltration mode, to leave 5 % of the MbA in the retentate; (3) total filtration time.

Equations

Concentration mode (constant flux, increasing TMP):

$$V_P(t) = Q_P\, t \tag{1}$$

$$J_v = \frac{Q_P}{A_m} = \frac{\mathrm{TMP}}{\mu\,(R_m + R_c(t))} \tag{2}$$

$$R_c(t) = \frac{R_c^*\, C_F(t)\, V_P(t)}{A_m} \tag{3}$$

$$C_F(t) = \frac{C_F^\circ\, V_F^\circ}{V_F^\circ - V_P(t)} \tag{4}$$

Diafiltration mode (constant volume, constant TMP), MbA balance on the feed tank:

$$V_F \frac{dC_{FP}}{dt} = \underbrace{Q_P\, C_{\mathrm{in}}}_{=0} - Q_P\, C_{FP} \quad \Rightarrow \quad t = -\frac{V_F}{Q_P} \ln\!\left(\frac{C_{FP}}{C_{FP}^\circ}\right) \tag{5}$$

Assumptions

- 50 % of the total biomass is retained by the membrane: $C_F^\circ = 0.5\times$ total biomass concentration in the feed.
- No data available for R_c^* or R_m; reasonable values assumed.
- Feed density and viscosity constant over the operation.
- MbA is not retained: its concentration is the same on both sides of the membrane, so the permeate leaves at C_{FP}.
- Diafiltration: buffer in at $Q_{\mathrm{in}} = Q_P$, so V_F is constant; fresh buffer carries no MbA, $C_{\mathrm{in}} = 0$.

Worked example

1. Concentration mode

Flux, from equation (2):

$$J_v = \frac{Q_P}{A_m} = \frac{1.5\,\mathrm{m^3/h}}{320\,\mathrm{m^2}} = 0.0046875\,\mathrm{m^3/(m^2\,h)}$$

Cake resistance at maximum TMP, rearranging equation (2):

$$\begin{aligned} J_v\,\mu\,(R_m + R_c(t)) = \mathrm{TMP}_{\max} \;\Rightarrow\; R_c(t) &= \frac{\mathrm{TMP}_{\max}}{J_v\,\mu} - R_m \\ &= \frac{(4\,\mathrm{bar})\,\left(1\times10^5\,\mathrm{Pa/bar}\right)}{(0.0046875\,\mathrm{m^3/(m^2\,h)})(0.001\,\mathrm{Pa\,s})} - 1\times10^9/\mathrm{m} \\ &= \boxed{8.433\times10^{10}/\mathrm{m}} \end{aligned}$$

Cake mass, rearranging equation (3):

$$C_F(t)\,V_P(t) = \frac{R_c(t)\,A_m}{R_c^*} = \frac{(8.433\times10^{10}/\mathrm{m})(320\,\mathrm{m}^2)}{1\times10^{11}} = \boxed{269.87\,\mathrm{kg}}$$

Feed concentration at that point, rearranging equation (4) as $C_F(t)V_F^\circ - C_F(t)V_P(t) = C_F^\circ V_F^\circ$:

$$C_F(t) = \frac{C_F^\circ V_F^\circ + C_F(t)\,V_P(t)}{V_F^\circ} = \frac{(27.834\,\mathrm{kg/m^3})(4.8\,\mathrm{m^3}) + 269.87\,\mathrm{kg}}{4.8\,\mathrm{m^3}} = \frac{403.47\,\mathrm{kg}}{4.8\,\mathrm{m^3}} = \boxed{84.06\,\mathrm{kg/m^3}}$$

The retained biomass has concentrated about threefold, from 27.83 to 84.06 kg/m^3. Permeate volume and time, from equations (4) and (1):

$$V_P(t) = \frac{C_F(t)\,V_F^\circ - C_F^\circ V_F^\circ}{C_F(t)} = \frac{403.47\,\mathrm{kg} - 133.60\,\mathrm{kg}}{84.06\,\mathrm{kg/m^3}} = 3.2105\,\mathrm{m^3}, \qquad t = \frac{V_P}{Q_P} = \frac{3.2105\,\mathrm{m^3}}{1.5\,\mathrm{m^3/h}} = \boxed{2.140\,\mathrm{h}}$$

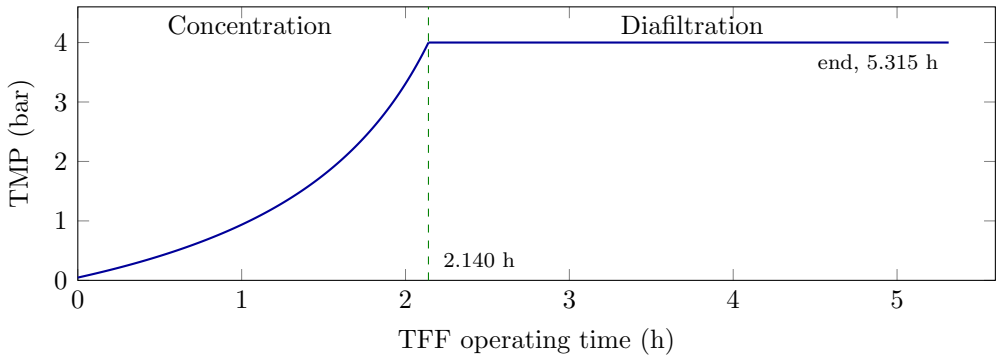
2. Diafiltration mode

Feed volume held from the end of stage 1 (taken at 2.14 h, as in the stream table): $V_F = V_F^\circ - Q_P t = 4.8 - (1.5)(2.14) = 1.59\,\mathrm{m^3}$, with $R_c = 8.433\times10^{10}/\mathrm{m}$ and TMP = 4 bar constant. Dividing the MbA balance (5) by $V_F C_{FP}$ and integrating:

$$\int_{C_{FP}^\circ}^{C_{FP}} \frac{dC_{FP}}{C_{FP}} = \int_0^t -\frac{Q_P}{V_F}\,dt \;\Rightarrow\; \ln\!\left(\frac{C_{FP}}{C_{FP}^\circ}\right) = -\frac{Q_P}{V_F}\,t \;\Rightarrow\; t = -\frac{1.59\,\mathrm{m^3}}{1.5\,\mathrm{m^3/h}}\ln(0.05) = \boxed{3.175\,\mathrm{h}}$$

3. Total and TMP profile

$$t_{\text{total}} = 2.140\,\mathrm{h} + 3.175\,\mathrm{h} = \boxed{5.315\,\mathrm{h}}$$



Stage 1 curve from equations (1)–(4): $\mathrm{TMP}(t) = \mu J_v\,[R_m + R_c^*\,C_F^\circ V_F^\circ\,Q_P t\,/\,(A_m(V_F^\circ - Q_P t))]$, rising from 0.05 bar to 4.00 bar at 2.140 h.

Check

- **Back-substitution:** TMP at $t = 2.14$ h from the profile equation is 3.998 bar, matching the stream table.
- **Buffer used in diafiltration:** $Q_P t = (1.5)(3.175) = 4.763\,\mathrm{m^3}$, or $-\ln(0.05) = 3.0$ diavolumes of V_F.
- **MbA recovery:** with MbA at equal concentration on both sides, stage 1 already sends $V_P/V_F^\circ = 3.21/4.8 = 66.9\,\%$ to the permeate. Diafiltration removes 95 % of the remaining 33.1 %, so overall recovery is $0.669 + 0.331(0.95) = 98.3\,\%$.